VALIKHUJAEV YAKHYO

Machine Learning | Deep Learning | Computer Vision Engineer

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Summary

Experienced computer vision engineer, with a passion for developing novel algorithms and systems that enable machines to see and understand the world around them. Proven ability to take complex problems and break them down into manageable parts, leading to rapid prototyping and successful product launches.

Proficient in self-supervised learning, supervised learning, unsupervised learning and transfer learning. Particularly interested in Generative AI, Medical AI, Explainable AI and MLOps.

Work Experience

Machine Learning Engineer *July 2022 – Sep 2023*

Pyler Co. Ltd *Seoul, South Korea*

- **Video-based Visual Content Moderation**: Build a Video Moderation Pipeline which helps to flag inappropriate video contents using latest state-of-the-art video recognition models, achieving a ~**10%** improvement of model accuracy.
- **Detection-based Visual Content Moderation**: Utilized segmentation and detection techniques to precisely detect unsuitable content for the brand safety. Implemented latest state-of-the-art models in terms of real-time speed and efficiency, improved the model precision and recall by ~**15%** by conducting **active learning** techniques. Build whole end-to-end pipeline using Kubeflow.
- **Classification-based Visual Content Moderation**: Leveraging multi-label and multi-head classification techniques improved the precision by ~**20%** while using self-supervised and supervised training approaches. This novel approach showcases the adaptability and efficacy of the model for *hard samples*.

AI Research Engineer *Nov 2020 – Jul 2022*

D-Meta Co. Ltd *Seoul, South Korea*

- **Slab text Recognition**: Developed and designed text detection and recognition model to efficiently recognize handwritten texts on slab metals using Spatial Transformer Networks and Sequential modeling. Built a whole pipeline from data pre-processing to training and evaluation of the model. Achieved **>90%** accuracy integrating state-of-the-art detection and recognition models for scene text images.
- **Automatic Number Plate Recognition**: Designed and developed ANPR model to accurately detect and recognize number plates. Leveraging active learning and synthetic image generation techniques improved the precision and recall by ~**15%**.
- **Car Damage Detection**: Built lightweight damage detection model and deployed it on an android device using torchscript. Improved the precision of the model by ~**10%** by tuning the model parameters.

Graduate Research Assistant *Sep 2018 – Feb 2021*

AI and SC lab at Gachon University *South Korea*

- **Computer Vision based Fire and Smoke Detection**
  - Lead the design and implementation of the dilated CNN architecture for improved feature extraction and recognition in images/videos. Carefully tuning and optimizing the model, achieved a high level of accuracy in fire and smoke detection, reducing false positives and having **1.5**x faster inference speed compared to the fastest counterpart.
  - Published all the findings and contributions in the paper titled "*Automatic Fire and Smoke Detection Method for Surveillance Systems Based on Dilated CNNs*".
- **O2O Industrial Project**
  - **Recommendation System**: Conducted a research on machine learning recommendation algorithms to achieve the best result on a given dataset for product recommendation. Improved model performance by conducting feature-engineering and feature-importance checks.
  - **Developing an application for remote control**: Built a remote controlling app for Android named *"7Harvest"* for rice peeling machine.
- **Model optimization for Edge device deployment**: Improved the FPS on Edge device (Raspberry PI 2) by leveraging hyper-parameter tuning and **quantization** on YOLOv3 detection model.
- **Decentralized Parking Occupancy Analysis Based on CNNs**: Parking Occupancy detection using dilated CNNs. Presented the paper at ISIS 2019 & ICBAKE 2019 Conference proceeding. Nominated with *"Best Presentation Award".*

Education

MSc in Computer Science *Sep 2018 – Feb 2021*

Gachon University *South Korea*

- Working as a TA and Researcher
- **Courses**: *Practical Algorithms, Advanced Artificial Intelligence, Advanced Algorithms, Big Data, Optimization Algorithms, Image Processing*.
- **Thesis**: *"An Efficient Fire and Smoke Detection Model for Surveillance Systems Based on Dilated CNNs"*.
- **GPA**: 4.01/4.5

BSc in Computer Science *Sep 2014 – Jun 2018*

Tashkent University of Information Technologies *Tashkent, Uzbekistan*

- Won the **2nd** place in the competition at the university by creating a desktop app for English Learners using C++ Builder.
- **GPA**: 85/100

Projects

Defect Detection

https://yakhyo.github.io/defect-det/

- Developed a detection model based on segmentation for detecting four types the defects of bolts on a conveyor belt.
- Implemented and compared the performances of several models and various types of loss functions.
- Conducted model Explainability assessment using **Grad CAMs** in order to check the inductive bias.

Road Crack Segmentation

https://yakhyo.github.io/crack-segmentation/

- Designed and developed an efficient, clean code project which uses UNet based model to effectively detect road cracks.
- Applied various loss functions and analyzed their performance. Evaluated the impact of the augmentation techniques to the performance of the model.

OCR Project

https://yakhyo.github.io/east-pytorch/

- Reproducing the results of the paper titled "EAST: An Efficient and Accurate Scene Text Detector"

- Created a clean and high-performance code base for training and evaluation stages.

ML Pipeline

https://github.com/yakhyo/ml-pipeline - private repo

- Designed a Machine Learning Pipeline from training to deployment. Implemented **Model Registry** for model artifacts using **WANDB**.
- Wrote highly efficient code using type checkers and style checkers. Created a **CI** with github actions which includes testing and linter checks.
- Deployed the model using **Flask** and Created an API for performing training and inference interactively. **Docker** and **Docker-compose** files are available to run the project by request.

Medical Image Segmentation

- Designing and Developing a clean-code based project in order to segment polyp from colonoscopy images
- At present working on improving the performance by utilizing transformer-based decoders.

Skills

Programming Languages	Python, C/C++, Java
ML Libraries and Frameworks	NumPy, Scikit-learn, PyTorch, PyTorch Lightning, Tensorflow, HuggingFace
MLOps Tools	Git/Github. Docker, Kubeflow, WANDB, MLFlow, Flask, Fast API
Project Management Tools	Jira, Asana, ClickUp, Notion, Slack
Soft Skills	Problem-solving, Teamwork, Communication, Time Management

Publications

Journal Article

Valikhujaev, Y., Abdusalomov, A., & Cho, Y. I. (2020). Automatic fire and smoke detection method for surveillance systems based on dilated CNNs. *Atmosphere*, *11*(11), 1241. https://doi.org/10.3390/atmos11111241

Honors

2019	**Best Presentation Award**	ISIS & ICBAKE 2019	Jeju, South Korea
2020	**Outstanding Paper Award**	Fire Investigation Society of Korea (FISK)	Seoul, South Korea

Languages

Uzbek	Native proficiency
English	Professional proficiency
Korean	Elementary proficiency
Russian	Intermediate proficiency